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A Report on:

" KhetiBot: Autonomous Pesticide Spraying for Nepal's Small Farms "

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Yours Sincerely,

Shlok Koirala

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ABSTRACT

This project presents the development of KhetiBot, an automated plant-detection and pesticide-spraying robot designed for small-scale farms in Nepal. The system aims to improve spraying efficiency, reduce manual labor, and minimize farmers' exposure to harmful chemicals. Initially, a physical prototype was developed using a Raspberry Pi 5, motor drivers, a relay module, and a DC water pump. However, power delivery and processing limitations led to a transition towards a simulation-based approach. The final system was developed using ROS 2, Blender, and Gazebo Classic. A custom YOLOv8n object detection model was trained using approximately 4,000 images collected from the Kathmandu University Panchkhal Campus premises, with tomato plants as the primary detection target. Upon detecting a target plant, the system automatically triggers a virtual spraying mechanism for three seconds while the robot follows a predefined trajectory. The simulation successfully demonstrated the integration of plant detection, robot movement, and automated spraying control. Although fully autonomous navigation and real-world field testing remain future work, KhetiBot provides a foundation for developing affordable precision agriculture solutions for Nepalese farmers.

Keywords: Precision Agriculture, YOLOv8n, ROS 2, Gazebo Classic, Plant Detection, Automated Spraying.

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ACRONYMS

CNN	: Convolutional Neural Network
DBSCAN	: Density-Based Spatial Clustering of Applications with Noise
ROS	: Robot Operating System
SLAM	: Simultaneous Localization and Mapping
URDF	: Unified Robot Description Format
YOLO	: You Only Look Once

CHAPTER - I

INTRODUCTION

1.1 Background of the Study

Agriculture is a vital sector in Nepal, engaging 62% of households and contributing 24% to the country's GDP. The majority of farmers operate on small, fragmented plots averaging 1.23 hectares, with 75% of farms below 1 hectare. Despite its importance, the sector faces challenges such as labor shortages due to urban migration, inefficient farming methods, and limited access to modern technology.

One of the most pressing issues is pesticide application, which is still predominantly manual. Farmers must carry and spray chemicals by hand, a method that is not only labor-intensive but also exposes them to serious health risks, including respiratory diseases and skin infections. The lack of protective equipment further worsens these dangers, especially in rural areas with limited healthcare access. Additionally, uneven and excessive pesticide use leads to environmental damage, affecting soil fertility, water sources, and biodiversity.

Nepal's diverse landscapes, ranging from plains to hilly and mountainous terrains, make large-scale mechanization difficult. The economic constraints of smallholder farmers further limit their ability to invest in advanced agricultural machinery. However, automation presents an opportunity to address these challenges effectively.

Pesticide-spraying robots offer a promising solution by ensuring precise, uniform spraying, reducing chemical exposure, and optimizing pesticide use. Given the constraints of small-scale farming, such technology must be affordable, accessible, and tailored to Nepalese agricultural conditions.

This project focused on simulating a plant-detection and targeted pesticide-spraying system and evaluated its feasibility in improving farming efficiency, safety, and sustainability. By bridging traditional farming with modern technology, it aimed to support smallholder farmers in addressing labor shortages, minimizing environmental impact, enabling more targeted spraying, reducing chemical exposure, and potentially lowering pesticide use.

The project initially pursued a physical prototype using a Raspberry Pi 5, motor driver, relay module, and DC water pump. Due to power delivery limitations encountered during hardware testing, development shifted to a simulation-based approach, as described in Chapters 3 and 6.

1.2 Problem Statement

Nepali farmers face significant challenges in pesticide application due to the reliance on manual labor, which is time-consuming, inefficient, and exposes them to harmful chemicals.

Uneven pesticide distribution leads to excessive chemical use, affecting both the environment and crop quality.

To address these issues, we propose a pesticide-spraying robot capable of detecting tomato plants and spraying pesticide on the detected plants . This system aims to reduce health risks, enhance spraying precision, minimize labor dependency, and contribute to the modernization of Nepalese agriculture. The present work demonstrates plant detection and spraying in simulation; fully autonomous navigation, including independently navigating fields and returning to base, is identified as a future extension.

1.3 Objectives

The primary objectives of this project were:

- To implement and evaluate CNN-based plant detection and a targeted spraying mechanism in a simulated agricultural environment.
- To identify key challenges and opportunities for autonomous agricultural solutions in Nepal, considering the unique constraints of small-scale farming, fragmented land holdings, and rugged terrains.

1.4 Motivation and Significance

Nepal's small-scale farmers face growing challenges due to labor shortages and increasing pest pressure, making modernized solutions essential. Automated pesticide sprayers can ease the labor burden, ensuring timely and efficient spraying on small plots where every plant impacts yield. Research on crop-adapted precision spraying systems has reported pesticide savings of 31% to 82% compared with conventional spraying methods . Such technologies have the potential to reduce costs for resource-limited farmers while minimizing environmental harm.

Small farms, located near diverse ecosystems, face higher pest risks, making effective pest control vital. Automated sprayers offer a safer alternative to manual spraying, reducing farmers' exposure to harmful chemicals. Additionally, their compact size makes small farms ideal for testing and deploying precision agriculture technologies, bridging traditional farming with modern advancements to ensure sustainability and resilience.

CHAPTER – II

RELATED WORK

The development of autonomous robotic systems for precision agriculture has gained significant attention in recent years, driven by the need to enhance resource efficiency, reduce environmental impact, and address labor shortages. This section reviews prior work relevant to our project, focusing on advancements in robotic sprayer systems and navigation techniques for agricultural robots.

Baltazar et al. (2021) presented a smart electric sprayer system designed for precision agriculture, emphasizing automated pesticide application [1]. Their work introduced a crop perception system that leverages a Support Vector Machine (SVM) classifier to estimate leaf density using image histograms derived from features such as Local Binary Patterns (LBP), vegetation index, average intensity, and hue. This density estimation drives a controller that adjusts air flow, water rate, and water density, optimizing pesticide delivery based on plant characteristics. The system, intended for robotic integration, achieved an accuracy of 80% to 85% in leaf density classification, demonstrating its potential to reduce chemical waste and improve spraying precision. Additionally, the authors contributed a publicly available dataset, enabling further research in this domain. While their approach excels in adapting spraying parameters to crop conditions, it does not address the navigation challenges of deploying such a system in complex agricultural environments, an aspect that lies outside the scope of our current work and is left as a future extension.

Ribeiro et al. (2022) explored a multifunctional agricultural robotic rover (R2A2) for weed control and fallen fruit collection in orchards, using computational simulation to streamline development [2]. Their system, simulated in CoppeliaSim and partially validated in field tests, integrates camera-based image recognition with AI algorithms to detect weeds and fruits, enabling localized herbicide spraying and gripper-based fruit collection. The rover's design emphasizes sustainability by minimizing herbicide use and automating labor-intensive tasks like collecting fallen peaches. Control algorithms manage locomotion and task execution, tested virtually in a simulated orchard environment. While field tests confirmed maneuverability, challenges such as wheel skidding under load were noted, indicating areas for improvement. This work complements Baltazar et al.'s spraying focus by adding fruit collection and leverages simulation to reduce development costs, though its navigation relies on simpler path-following compared to more advanced techniques.

Jiang and Ahamed (2023) focused on navigation for an autonomous spraying robot tailored to orchard operations, particularly in traditional Japanese orchards with low-height trees and dense canopies [3]. Their system employs a 2D LiDAR (Hokuyo UST-10LX) as the primary sensor for tree trunk detection, circumventing the limitations of GNSS-based navigation in environments where satellite signals are obstructed. The navigation pipeline uses

Density-Based Spatial Clustering of Applications with Noise (DBSCAN) to identify tree trunk clusters from LiDAR point clouds, followed by K-Means clustering to locate trunk centroids, and Random Sample Consensus (RANSAC) to fit a navigation path through the tree rows. Implemented on a tracked robot powered by an NVIDIA Jetson Nano and integrated via the Robot Operating System (ROS), their solution achieved navigation accuracy with Root Mean Square Errors (RMSE) ranging from 9.4 cm to 18.6 cm in an artificial orchard setting. This GNSS-free approach highlights the efficacy of LiDAR-based navigation for structured agricultural layouts, though it lacks the spraying optimization features addressed by Baltazar et al.

The reviewed studies provide a foundation for the development of KhetiBot by demonstrating the potential of machine learning, robotic systems, and simulation technologies in precision agriculture. Inspired by the targeted spraying approach of Baltazar et al. and the simulation-based development approach of Ribeiro et al., this project focuses on integrating vision-based plant detection with automated spraying control. Camera-based Visual SLAM was also explored as a possible navigation approach; however, autonomous mapping and localization were not integrated into the final system. Instead, the robot follows predefined trajectories within the ROS 2 simulation environment. Therefore, the primary contribution of this project is the development and simulation of a plant-detection and automated spraying pipeline using YOLOv8n, ROS 2, and Gazebo Classic.

CHAPTER – III

METHODOLOGY

This project involved the development of a plant-detection and automated spraying robot for potential application in small-scale agricultural fields. The system uses a custom-trained YOLOv8n object detection model to identify tomato plants and trigger spraying operations. Initially, physical hardware prototyping was attempted; however, technical limitations related to power delivery and processing performance led to a transition towards simulation-based development. The final system was implemented using ROS 2 and Gazebo Classic, with a virtual environment created in Blender to test robot movement, plant detection, and spraying control.

3.1 Dataset Preparation and Object Detection Model

To establish the perception foundation for the system, dataset collection and model training were prioritized as the initial step. Around 4,000 photographs were captured across the Panchkhal Campus premises to ensure the data reflected real-world environmental conditions. These images were manually annotated into several distinct classes, including road, ground, grass, tomato plant, human, buildings, and others.

Among these, the tomato plant class served as the primary target to trigger the automated spraying mechanism. The remaining background classes were incorporated to assist the model in effectively distinguishing target plants from surrounding elements and minimizing false detections. A YOLOv8n object detection model was subsequently trained on this curated dataset, which was partitioned into dedicated training and validation subsets to ensure robust learning and evaluation.

3.2 System Development Approach

Following the training of the YOLOv8n perception model, the project's engineering methodology encompassed both a physical hardware prototyping phase and a simulation-driven validation phase. Initially, a physical prototype was constructed using a Raspberry Pi 5 as the central processing unit, interfaced with a motor driver for differential-drive locomotion, a relay switch, and a DC water pump for pesticide spraying, as detailed in the hardware architecture and circuit schematics.

While preliminary bench tests were successful, hardware testing revealed severe power delivery constraints, specifically undervoltage throttling and unexpected shutdowns caused by the high current draw of the Raspberry Pi 5 under AI inference loads. To overcome these physical limitations and continue development safely, the project transitioned to a simulation-driven approach.

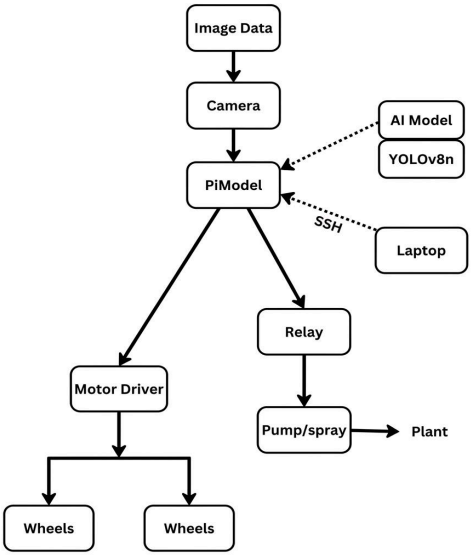


Figure 1: Hardware Architecture

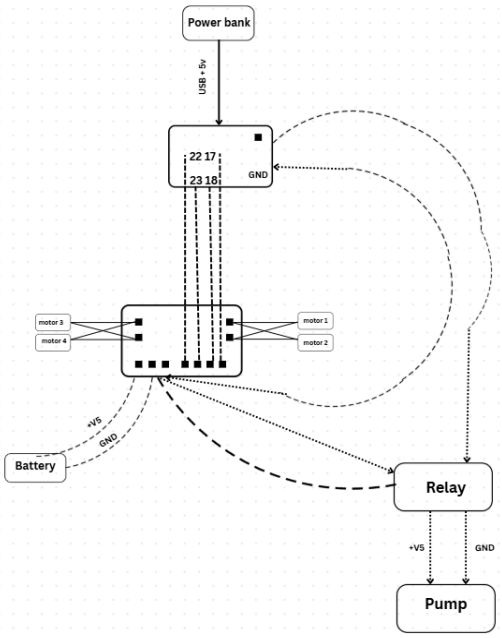


Figure 2: Circuit Diagram

The core perception model and control logic were deployed within a digital twin environment using ROS 2, Blender, and Gazebo Classic, enabling reliable testing of robot motion, computer vision, and targeted spraying operations.

3.3 Environment Creation

To simulate and validate KhetiBot's functionality prior to real-world deployment, we developed a high-fidelity virtual environment using Blender. The environment was modeled to replicate the 3D layout of the Panchkhal Campus premises, incorporating realistic terrain features, architectural structures, and natural obstacles. This digital replica allowed us to create a controlled and repeatable simulation platform where KhetiBot's core capabilities, such as plant detection, robot motion, and targeted spraying operations, could be tested and refined. By leveraging this virtual environment, we were able to identify potential issues early and fine-tune robot behavior without the constraints and risks associated with physical field trials.

3.3.1 Designing the 3D Environment in Blender

The virtual environment was developed using Blender to replicate the Panchkhal Campus layout. Key landscape elements including grassy fields, academic buildings, pavements, trees, and natural as well as man-made obstacles were modeled. Particular attention was given to maintaining realistic proportions and spatial relationships based on actual site dimensions.

The 3D scene was enhanced with textures and surface details to improve visual realism, which is essential for validating vision-based plant detection. Once the modeling was complete, the scene was exported in .dae or .glb format to ensure compatibility with Gazebo Classic, the simulation platform used for testing robot behavior in a controlled setting.

The final map dimensions are approximately 59.4 meters (X-axis) by 37 meters (Y-axis) with a vertical height of 9 meters (Z-axis), providing a realistic scale for spraying simulations.



Figure 3: 3D Environment

3.3.2 Designing Robot

The robot was designed as a four-wheeled skid-steer mobile robot, a configuration chosen for its simplicity, maneuverability and stability on uneven ground. The two wheels on the left side and the two wheels on the right side were driven together as pairs. The robot moved forward or backward when both sides turned at the same speed, and it turned when the two sides turned at different speeds. The robot's structure was described using URDF and Xacro files in ROS 2. The dummy link served as the robot's base frame, and the chassis (body_link) was attached to it. Each of the four wheels (wheel1_link to wheel4_link) was modeled as a cylinder and connected to the base through its own joint (wheel1_joint to wheel4_joint), allowing it to rotate freely. The chassis and wheels were given distinct colors in Gazebo to make them easy to tell apart during simulation.

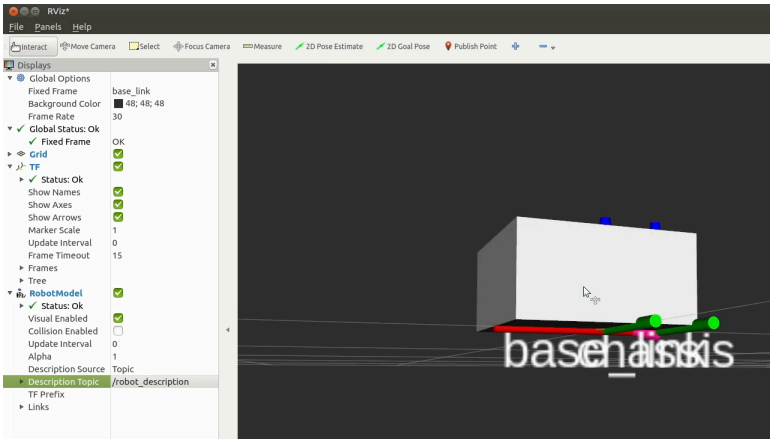


Figure 4: URDF Robot Design

To simulate realistic behavior, all robot links included visual, collision, and inertia properties, with the inertia parameters calculated using predefined macros for accuracy and ease. Motion was produced by the Gazebo differential-drive plugin, configured for two wheel

pairs, which converted velocity commands into left and right wheel motion. This modular, standards-compliant design supported effective simulation and control within ROS 2 and allowed the camera and spraying module to be added to the same model.

TF Tree Overview

The TF (Transform) tree shows how all parts of the robot are connected in 3D space using coordinate frames. In our setup, the root frame is odom, which connects to the dummy frame (the robot's base), and from there to the wheels and main body. This structure allows ROS 2 to keep track of how each part of the robot moves over time.

In our robot, the TF tree begins at the odom frame, which represents the odometry reference frame from which the robot's motion is measured. This connects to the dummy frame (our robot's base), which then branches out to each component, including the four wheels (wheel1_link to wheel4_link) and body_link. These relationships define how the robot's parts are positioned relative to each other. The transform from odom to dummy and the wheel transforms were published by the Gazebo drive plugin as the robot moved.

The data from the TF tree is published over the /tf topic. This topic provides the transformations between frames, so any tool or node can subscribe to it and get up-to-date information about the robot's structure and pose. Visual tools like RViz2 use this tree to visualize the robot correctly, while navigation algorithms depend on it to perform tasks like localization, path planning, and obstacle avoidance. A well-structured TF tree ensures the robot's internal coordinate frames remain consistent and accurate throughout its operation.

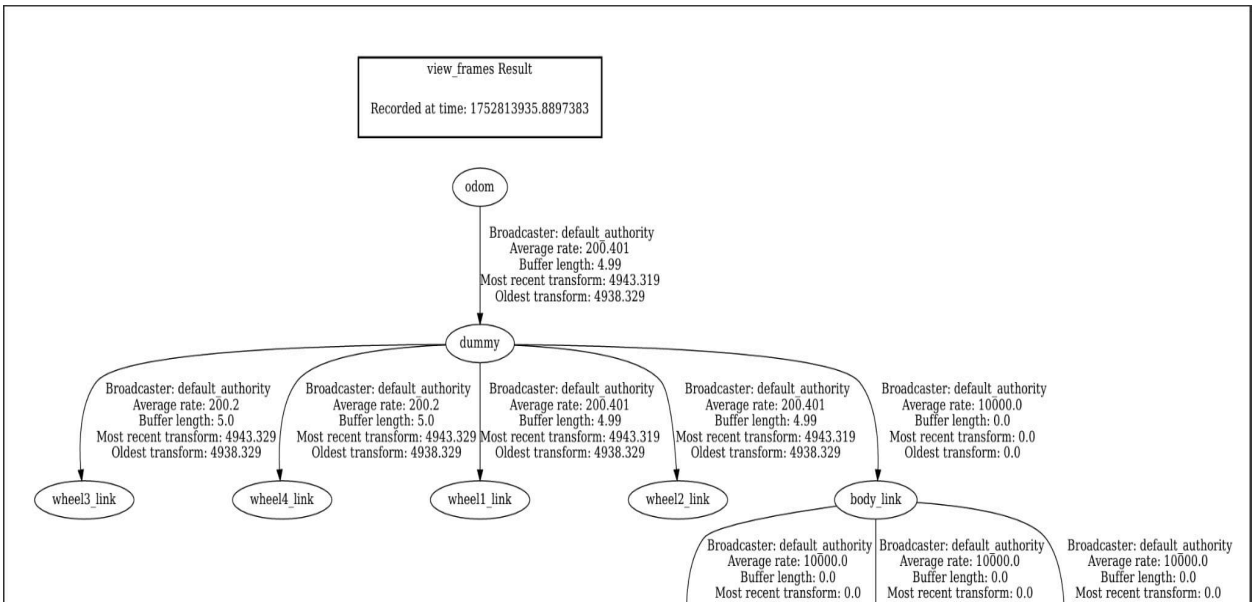


Figure 5: TF Tree architecture

3.3.3 Integrating the Spraying Robot Model

After configuring the environment, we introduced our custom-built spraying robot model into the simulation. The robot was defined using URDF (Unified Robot Description Format) and SDF (Simulation Description Format), which described its physical properties, joint configurations, sensors (like cameras), and spraying mechanisms. The integration also included control plugins to simulate differential drive movement and sensor feedback. Once placed into the map, the robot was tested for mobility, sensor range, and interaction with the environment to ensure all components responded correctly to the simulation engine.

3.3.4 Importing and Configuring the Simulation in Gazebo Classic

The exported Blender environment was imported into Gazebo Classic, a widely used robotics simulator that supports dynamic physics and sensor modeling. Inside Gazebo, we fine-tuned the lighting conditions, terrain friction, collision parameters, and gravitational settings to resemble real-world conditions. Additional objects such as barriers, vegetation, and field markers were placed within the map to add realistic scene complexity. These elements helped simulate the conditions the spraying robot would face in an actual agricultural or campus-like field environment.

3.4 Robot Motion and Spraying

In the simulation environment, the robot's movement and execution are governed by a differential-drive control architecture. Rather than relying on dynamic autonomous mapping (such as Visual SLAM which is deferred to future work), the robot navigates via a pre-defined path or velocity commands (cmd_vel) issued through the ROS 2 control framework.

The robot's movement is controlled through a four-wheeled skid-steer configuration, allowing linear and angular motion based on velocity commands. At the same time, the camera feed is processed by the YOLOv8n object detection model to identify tomato plants. When a target plant is detected, the control system activates a virtual relay and spraying mechanism for a fixed duration of three seconds. After the spraying sequence is completed, the virtual spraying mechanism is deactivated and the robot continues along its predefined trajectory.

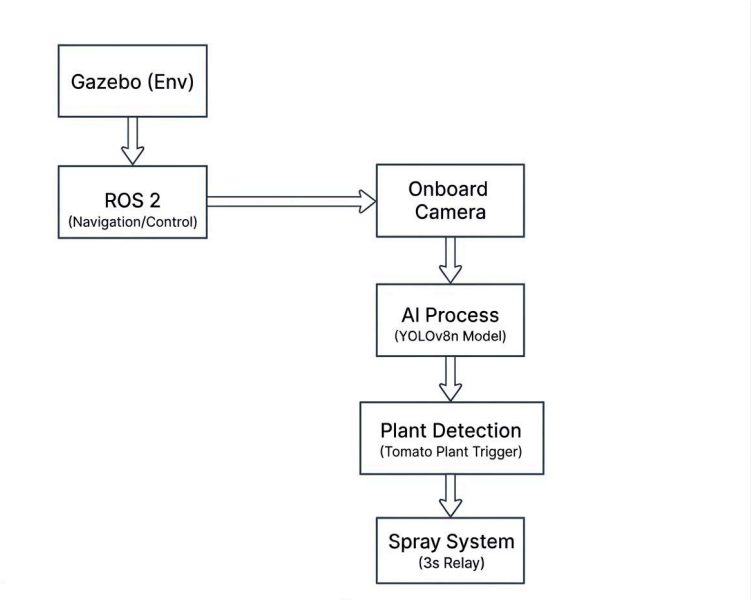


Figure 6: Simulation Architecture

3.5 Testing Detection and Spraying Functionalities

Following the integration of the object detection model, a series of simulation tests were conducted to evaluate the robot's ability to detect tomato plants and execute targeted spraying tasks in the Panchkhal Campus environment.

The spraying mechanism was tested in conjunction with the object detection model (YOLOv8n), which continuously analyzed the visual feed to identify tomato plants. Upon detection, the control system triggered the spraying module to activate for a fixed duration of three seconds, simulating the spraying process. The effectiveness of this action was assessed by monitoring the timing of the spray command and its alignment with the detected plant positions.

The system's operation was observed through the simulation output, including the detected plant positions and the corresponding spraying commands. The terminal output was used to verify whether the spraying command was triggered following target detection. These tests provided a basic demonstration of the interaction between the object detection model and the virtual spraying mechanism. Further quantitative evaluation and real-world testing are required to assess the system's overall performance under practical agricultural conditions.

CHAPTER – IV

TOOLS, TECHNOLOGIES, AND HARDWARE USED

The system requirements are tailored to support the development, simulation, AI training, and evaluation of a plant-detection and pesticide-spraying robot in a virtual agricultural environment representative of Nepalese farmlands. These requirements ensure that the robot can detect target plants and make spraying decisions efficiently within a ROS-Gazebo simulation setup.

Software Used:

- **3D Modeling Tool: Blender –** for designing the 3D campus environment, including buildings, terrain, vegetation, and field obstacles.
- **Robot Modeling:** URDF and Xacro – for defining the robot's chassis, four wheels, and joints in ROS 2.
- **Simulation Platform:** Gazebo Classic – for simulating robot motion, sensor feedback, and environmental interactions.
- **Robot Middleware:** ROS 2 (Robot Operating System) – for integrating sensors, control logic, AI modules, and data communication.
- **Programming Language:** Python – for AI model development and control logic implementation.

Hardware Used

- **Workstation and Simulation Environment:**
    - **Processor:** A multi-core processor (Intel core i7 / AMD Ryzen 9) was used to support computational tasks and run the simulation nodes.
    - **Memory:** System memory was used to handle the simulation environment, sensor data, and AI processing tasks.
    - **Graphics:** A dedicated GPU (NVIDIA GeForce RTX 3080 with 10 GB GDDR6X memory) was used to support the training and inference of the YOLOv8n perception model.
    - **Storage:** SSD storage was used to store simulation assets, model checkpoints, and project files.
- **Prototyping and Robot Hardware:**
    - **Processing Unit:** A Raspberry Pi 5 was used as the central onboard controller during the physical prototype development phase.
    - **Locomotion and Actuation:** A motor driver was used to control the four-wheeled skid-steer drive system. A relay module and DC water pump were also included in the prototype for the spraying mechanism.

- **Perception Hardware:** An onboard camera module was used for video capture and object detection.

CHAPTER – V

RESULTS AND OUTPUTS

The simulation demonstrated the core functionalities of KhetiBot, particularly plant detection, robot movement, and automated spraying control within the virtual environment.

The YOLOv8n-based object detection model identified tomato plants through bounding boxes and class labels. When a target plant was detected, the control system generated a spraying command to activate the virtual spraying mechanism for three seconds. The successful triggering of this command was observed through the terminal output, as shown in Figure 7.

During the simulation, the robot followed a predefined trajectory through the environment while the detection model processed the camera feed. This demonstrated the integration of the perception and control components within the ROS 2 simulation framework. However, these results represent simulation-based functionality rather than validation through actual pesticide spraying or real-world agricultural field testing.

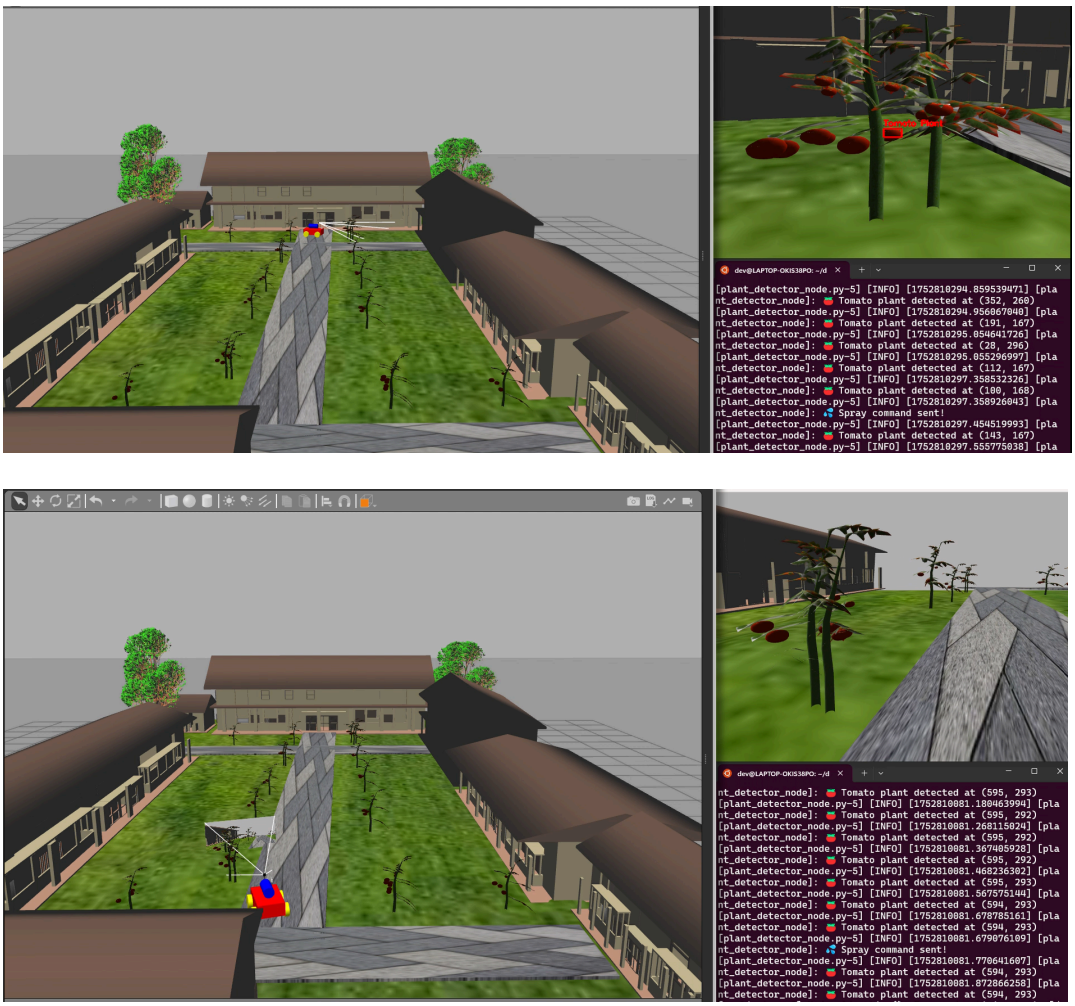


Figure 7: Results

5.1 YOLOv8n Model Evaluation

The performance of the trained YOLOv8n object detection model was evaluated using precision, recall, and mean Average Precision (mAP) at an IoU threshold of 0.50. The model achieved a precision of 81.04 %, a recall of 79.32%, and an mAP@50 of 84.13%.

These results indicate that the model was able to identify the target objects with a reasonable level of detection performance. The precision score reflects the proportion of correct detections among all predicted detections, while the recall score represents the proportion of actual objects successfully identified by the model. The mAP@50 score provides an overall measure of detection accuracy at an IoU threshold of 0.50.

Overall, the evaluation results demonstrate the potential of the YOLOv8n model for plant detection in the KhetiBot system. The model's detection output was further integrated with

the ROS 2 control system to trigger the virtual spraying mechanism upon detecting tomato plants.

CHAPTER VI

PROBLEMS FACED AND LIMITATIONS

6.1 Power Usage

The Raspberry Pi 5 model, although computationally more powerful than its predecessors, has significantly higher current requirements. During hardware testing, it was observed that standard 5V power sources, including high-capacity power banks, were insufficient to meet the current demand. This caused frequent under-voltage warnings, performance throttling, and unexpected shutdowns. These limitations in power delivery negatively impacted the stability and real-time performance of the system.

6.2 Throughput Constraints

The power supply limitations directly impacted overall processing throughput on the physical prototype. Executing real-time computer vision inference on the edge processor (Raspberry Pi 5) under constrained power delivery introduced significant processing latency. Under high-load scenarios, video frame acquisition and target processing experienced severe latency spikes and frame drops. This delay directly affected motor control and relay responsiveness, reinforcing the necessity of transitioning development to a high-performance host workstation (RTX 4080) to reliably evaluate the custom YOLOv8n plant perception pipeline within the Gazebo simulation environment.

6.3 System Limitations

- **Detection Limitations:** The object detection model may occasionally misclassify distant, partially visible, or shadowed plants. Such detection errors can affect the spraying decisions made by the control system. Further testing and model optimization are required to improve detection reliability under different environmental conditions.

- **Semi-autonomous Nature:** The system operates in a semi-autonomous mode, relying on predefined logic and lacking adaptive intelligence for dynamic environments. This makes it less reliable in unfamiliar or cluttered settings.
- **Navigation:** Visual SLAM was explored but not integrated, so the robot does not build maps or localize itself autonomously and follows a predefined path.
- **Reliability:** Hardware limitations and lack of robust error-handling mechanisms mean the system is susceptible to sudden failures, especially in outdoor or high-mobility environments.

CHAPTER – VII

CONCLUSION AND FUTURE ENHANCEMENTS

The development of a virtual simulation environment for KhetiBot enabled testing and validation of its design and core functionalities before real-world deployment. By leveraging Blender for 3D modeling and Gazebo Classic for physics-based simulation, we created a simulated model of the Panchkhal Campus and a functional differential-drive spraying robot. Integration of vision-based object detection using YOLOv8n further enabled the robot to identify plant targets and trigger spraying tasks within the simulation. This modular approach using ROS 2 ensured flexibility, scalability, and a foundation for future upgrades.

The system has not yet been tested in actual agricultural fields, and autonomous navigation was not achieved in this work; Visual SLAM was explored but not integrated. Despite the successful outcomes in simulation, several areas remain for enhancement. Future work will focus on:

- **Throughput Optimization:** Offload heavy processing (like DNN inference) to a remote Jetson Nano or GPU server over Wi-Fi, or use a Google Coral USB Accelerator for edge TPU-accelerated detection.
- **Proper Task Scheduling:** Use async task queues or separate microcontrollers (like Arduino) to offload motor control logic from the Pi.
- **Autonomy Enhancements:** Integrate a GPS + compass module for basic navigation.
- **Real-world Testing:** Deploy and test the robot in agricultural fields to validate system robustness under variable conditions.
- **Advanced Navigation:** Integrate Visual SLAM and multi-sensor fusion for improved localization, mapping, and dynamic obstacle avoidance.
- **Adaptive Spraying:** Use depth sensing and AI to optimize chemical usage based on plant size or health.
- **Multi-robot Coordination:** Enable cooperative behavior and workload distribution for large-scale field coverage.
- **Cloud-based Monitoring:** Allow remote access to robot data, live video feeds, and performance analytics.

These enhancements will further improve the robot's efficiency, adaptability, and usability in precision agriculture applications.

REFERENCES

[1] Baltazar, A. R., Santos, F. N. D., Moreira, A. P., Valente, A., & Cunha, J. B. (2021). Smarter robotic sprayer system for precision agriculture. *Electronics, 10*(17), 2061.

[2] Ribeiro, J. P. L., Gaspar, P. D., Soares, V. N. G. J., & Caldeira, J. M. L. P. (2022). Computational simulation of an agricultural robotic rover for weed control and fallen fruit collection: Algorithms for image detection and recognition and systems control, regulation, and command. *Electronics, 11*(5), 790.

[3] Jiang, A., & Ahamed, T. (2023). Navigation of an autonomous spraying robot for orchard operations using LiDAR for tree trunk detection. *Sensors, 23*(10), 4808.

[4] National Statistics Office. (2023). *National Sample Census of Agriculture 2021/22: National Report*. Government of Nepal, Kathmandu, Nepal.

[5] Ministry of Finance. (2024). *Economic Survey 2023/24*. Government of Nepal, Kathmandu, Nepal.

[6] Doruchowski, G., Balsari, P., Marucco, P., van de Zande, J. C., & Wenneker, M. (2012). Crop adapted spray application (CASA): Precise and safe plant protection in fruit growing. *Aspects of Applied Biology, 114*, 129–136.